

Rudra Arora

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SUMMARY

Third-year Computer Science student at York University who builds full-stack applications end to end, from React frontends and REST APIs to backend services, data pipelines, and LLM integrations. Comfortable picking up unfamiliar codebases, integrating third-party and AI APIs, and validating what comes back rather than trusting it. Test-first by habit. Three years leading a distributed student team alongside the technical work.

TECHNICAL SKILLS

Languages: JavaScript, Python, Java, SQL, HTML, CSS, TypeScript

Frontend: React, REST API Integration, Streamlit, Plotly

Backend & Data: Spring Boot, FastAPI, Node.js/Express, RESTful API design, concurrency

Databases: PostgreSQL, SQL query design

AI & ML: LLM API integration, prompt engineering, AI prototyping and POC development, scikit-learn, LightGBM, PyTorch, pandas, NumPy, feature engineering, model evaluation and validation

Tools & Practices: Git, pytest, JUnit, Agile, SDLC, Test-Driven Development, Docker

EDUCATION

York University

Winter 2024-Present

- Pursuing Bachelor of Science in Computer Science
- Enrolled in York University's Co-operative Education Program

PROJECTS

Job Market Salary Predictor (Python, scikit-learn, FastAPI, PostgreSQL, pytest)

- Engineered a feature pipeline over 19,000+ job postings covering target encoding, job categorization, and location parsing, then rebuilt it as a two-stage TF-IDF into gradient-boosting stack, raising accuracy from 54.6% to 79.8% against a 47.8% baseline.
- Compared RandomForest and HistGradientBoosting using stratified splits and 5-fold cross-validation, and validated the final model on unseen employers to confirm it generalized rather than memorized.
- Served the model as a FastAPI microservice backed by PostgreSQL and a 57-test pytest suite covering inference logic and API behaviour.

AI-Powered Consultation Booking Platform (Java, Spring Boot, PostgreSQL, Gemini API, JavaScript, HTML, CSS)

- Built the full stack of a consultation booking system: frontend, REST API, and PostgreSQL schema, containerized with Docker.
- Designed the complete appointment lifecycle with automated state-transition validation, acting as a programmatic quality gate that blocked invalid booking progressions at the service layer instead of relying on frontend checks.
- Integrated the Google Gemini API to embed an assistant inside the booking flow itself, answering user questions in real time at the point of friction rather than as a separate chatbot.

Full-Stack Chess Engine (Java, Springboot, REST, JS)

- Built a user-facing chess app with 3 game modes and a server-side AI using minimax with alpha-beta pruning (depth 3), cutting the search space ~70% versus brute-force
- Enforced complete rule validation server-side (castling, en passant, promotion, check/checkmate/stalemate), fully decoupling game logic from the frontend
- Managed concurrent game sessions with ConcurrentHashMap and validated game logic with unit tests across isolated components.

EXPERIENCE

Student Success Mentor Lead - YorkU Libraries

Fall 2024-Present

- Lead and coordinate a team of 5 mentors across 3 campuses, implementing structured scheduling workflows to reduce coordination delays and improve execution efficiency.
- Standardized cross-platform content scheduling workflows, contributing to a **30% increase** in audience growth.
- Plan and execute **2 multi-campus events per month**, driving a **40% increase in event sign-ups** through targeted social media promotion and digital outreach strategies.
- Manage WordPress updates and multi-platform content distribution (Instagram, Twitter, YouTube), leveraging scheduled and manual posting strategies while tracking engagement metrics to refine outreach performance.

Study Space Assistant - YorkU Libraries**May 2026 - Present**

- Supported day-to-day library operations across 5 campus locations, coordinating shipping logistics and resource distribution between facilities
- Assisted students with printing, photocopying, scanning, and computer troubleshooting, resolving technical and logistical issues on the spot
- Provided on-site event support during library-hosted events, managing space setup, equipment, and student-facing coordination